

SECR2043 OPERATING SYSTEMS

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 Section : Section 3

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write **"No output"** instead.

1.	Display your current working directory.	
	Command	Output
	pwd	<pre>syahmi@secr2043:~\$ pwd /home/syahmi</pre>
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	Output
	ls -a /	<pre>syahmi@secr2043:~\$ ls -a / . boot lib media root snap usr .. dev lib.usr-is-merged mnt run srv var bin etc lib64 opt sbin sys bin.usr-is-merged home lost+found proc sbin.usr-is-merged tmp</pre>
3.	Create a new directory named "os_lab" in your home directory.	
	Command	Output
	mkdir os_lab	<pre>syahmi@secr2043:~\$ mkdir os_lab syahmi@secr2043:~\$ ls os_lab</pre>
4.	Change your current working directory to "os_lab".	
	Command	Output
	cd os_lab	<pre>syahmi@secr2043:~\$ cd os_lab syahmi@secr2043:~/os_lab\$ </pre>
5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	Command	Output
	touch hello.txt echo "Hello, world!" > hello.txt	<pre>syahmi@secr2043:~/os_lab\$ touch hello.txt syahmi@secr2043:~/os_lab\$ echo "Hello, world!" > hello.txt</pre>

6.	Display the content of "hello.txt".	
	Command	Output
	cat hello.txt	syahmi@secr2043:~/os_lab\$ cat hello.txt Hello, world!
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.	
	Command	Output
	cp hello.txt hello_copy.txt	syahmi@secr2043:~/os_lab\$ cp hello.txt hello_copy.txt syahmi@secr2043:~/os_lab\$ ls -R .: hello.txt hello_copy.txt syahmi@secr2043:~/os_lab\$ cat hello_copy.txt Hello, world!
8.	Rename "hello_copy.txt" to "hello_bak.txt".	
	Command	Output
	mv hello_copy.txt hello_bak.txt	syahmi@secr2043:~/os_lab\$ mv hello_copy.txt hello_bak.txt syahmi@secr2043:~/os_lab\$ ls -R .: hello.txt hello_bak.txt
9.	Delete "hello.txt".	
	Command	Output
	rm hello.txt	syahmi@secr2043:~/os_lab\$ rm hello.txt syahmi@secr2043:~/os_lab\$ ls -R .: hello_bak.txt
10.	Display your home directory with additional information, together with subdirectories contents.	
	Command	Output
	ls -R ~	syahmi@secr2043:~/os_lab\$ ls -R ~ /home/syahmi: os_lab /home/syahmi/os_lab: hello_bak.txt
Total Mark:		

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

For each question, write down the command and the output (if any) in the answer sheet.

1.	Create an empty file named "hello.txt" in the current working directory.	
	Command	Output
	touch hello.txt	<pre>syahmi@secr2043:~/os_lab\$ touch hello.txt syahmi@secr2043:~/os_lab\$ ls hello.txt hello_bak.txt</pre>
2.	Write the text "Hello World" to "hello.txt" using echo.	
	Command	Output
	echo "Hello, world!" > hello.txt	<pre>syahmi@secr2043:~/os_lab\$ echo "Hello, world!" > hello.txt</pre>
3.	Append the text "This is a test" to "hello.txt" using echo.	
	Command	Output
	echo "This is a test" >> hello.txt	<pre>syahmi@secr2043:~/os_lab\$ echo "This is a test" >> hello.txt</pre>
4.	Display the contents of "hello.txt" using cat.	
	Command	Output
	cat hello.txt	<pre>syahmi@secr2043:~/os_lab\$ cat hello.txt Hello, world! This is a test</pre>
5.	Create another file named "bye.txt" with the text "Goodbye World" using echo.	
	Command	Output
	touch bye.txt echo "Goodbye World" > bye.txt	<pre>syahmi@secr2043:~/os_lab\$ touch bye.txt syahmi@secr2043:~/os_lab\$ ls bye.txt hello.txt hello_bak.txt syahmi@secr2043:~/os_lab\$ echo "Goodbye World" > bye.txt</pre>
6.	Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.	
	Command	Output
	cat hello.txt bye.txt	<pre>syahmi@secr2043:~/os_lab\$ cat hello.txt bye.txt Hello, world! This is a test Goodbye World</pre>
7.	Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.	
	Command	Output
	cat hello.txt bye.txt > greeting.txt	<pre>syahmi@secr2043:~/os_lab\$ cat hello.txt bye.txt > greeting.txt</pre>

8.	Display the contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	<pre>syahmi@secr2043:~/os_lab\$ cat greeting.txt Hello, world! This is a test Goodbye World</pre>
9.	Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.	
	Command	Output
	nano greeting.txt	<pre>syahmi@secr2043:~/os_lab\$ nano greeting.txt</pre> 
10.	Display the modified contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	<pre>syahmi@secr2043:~/os_lab\$ cat greeting.txt Hello, Syahmi Faris! This is a test Goodbye World</pre>
	Total Mark:	

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc.
For each question, write down the command and the output (if any) in the answer sheet.

1.	Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".	
	Command	Output
	nano hello.c	<pre>syahmi@secr2043:~/os_lab\$ nano hello.c</pre>
2.	Compile the program "hello.c" using gcc and generate an executable file named "hello".	
	Command	Output
	gcc hello.c -o hello	<pre>syahmi@secr2043:~/os_lab\$ gcc hello.c -o hello</pre>
3.	Run the executable file "hello" and display its output.	
	Command	Output
	./hello	<pre>syahmi@secr2043:~/os_lab\$./hello Hello, World!</pre>
4.	Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".	
	Command	Output
	nano sum.c	<pre>syahmi@secr2043:~/os_lab\$ nano sum.c</pre>
5.	Compile the program "sum.c" using gcc and generate an executable file named "sum".	
	Command	Output
	gcc sum.c -o sum	<pre>syahmi@secr2043:~/os_lab\$ gcc sum.c -o sum</pre>
6.	Run the executable file "sum" with 10 and 20 as arguments and display its output.	
	Command	Output
	./sum 10 20	<pre>syahmi@secr2043:~/os_lab\$./sum 10 20 10 + 20 = 30</pre>
7.	Run the executable file "sum" with -5 and 15 as arguments and display its output.	
	Command	Output
	./sum -5 15	<pre>syahmi@secr2043:~/os_lab\$./sum -5 15 -5 + 15 = 10</pre>

8.	Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".	
	Command	Output
	nano echo.c	syahmi@secr2043:~/os_lab\$ nano echo.c
9.	Compile the program "echo.c" using gcc and generate an executable file named "echo".	
	Command	Output
	gcc echo.c -o echo	syahmi@secr2043:~/os_lab\$ gcc echo.c -o echo
10.	Run the executable file "echo", type "Hello World" as input and display its output.	
	Command	Output
	./echo	syahmi@secr2043:~/os_lab\$./echo Enter a line of text: Hello World You entered: Hello World.
	Total Mark:	